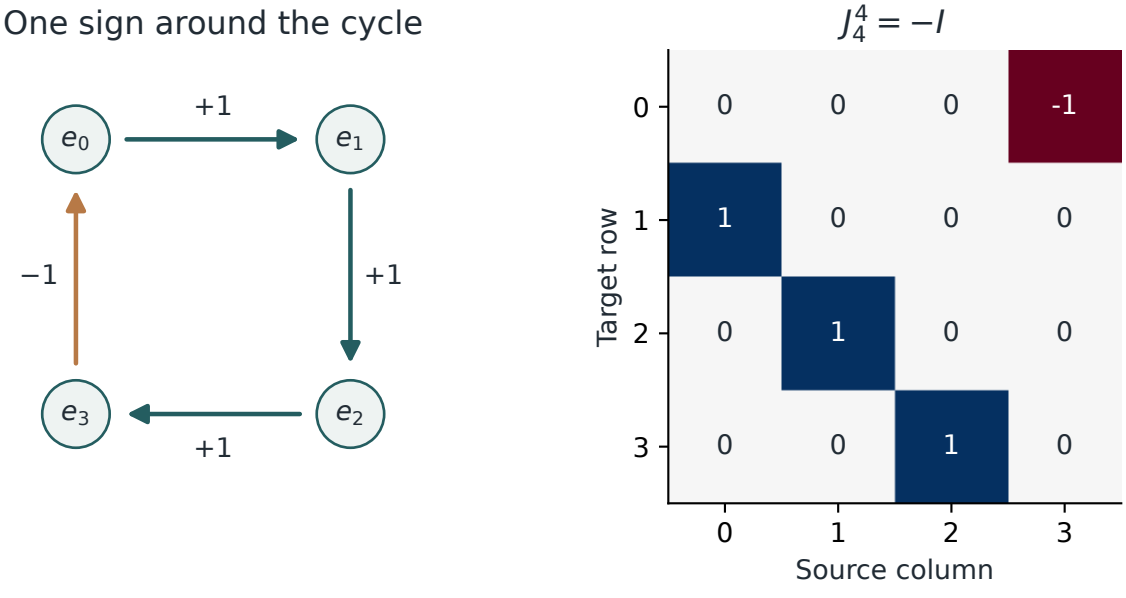
One sign around the cycle
e_0
e_1
e_2
e_3
+1
+1
+1
−1
$J_4^4 = -I$
Target row
Source column
0
1
2
3
0 0 0 -1
1 0 0 0
0 1 0 0
0 0 1 0